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# Price Discovery and Volatility Transmission in Currency Spot and Futures Markets in India: An Empirical Analysis

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## Abstract

This article investigates the price discovery and volatility transmission between currency spot and futures markets in India. Daily closing currency spot and futures prices of United States dollar/Indian Rupee (USD/INR), Japanese Yen/Indian Rupee (JPY/INR), Great Britain Pound/Indian Rupee (GBP/INR) and Euro/Indian Rupee (EUR/INR) from 1 March 2010 to 20 January 2015 were used. Both long- and short-run relationship as well as volatility transmission was investigated by the use of Engle Granger co-integration test, error correction mechanism, causality test and the bivariate GARCH model. The co-integration results proved that there is a long-run relationship between currency spot and futures prices. The study finds that there is a unidirectional causal relationship from currency spot to futures prices of JPY/INR, GBP/INR and EUR/INR. Further, results conclude the bidirectional causal relationship between USD/INR currency spot and futures prices. The result further shows that the spot market adjusts to new information faster than futures market suggesting that spot price leads the futures price and contributes largely to price discovery. Finally, the results show that there is unidirectional volatility transmission from currency spot to futures prices of JPY/INR, GBP/INR and EUR/INR and bidirectional spillover between currency spot and futures prices of USD/INR. Based on the findings, relevant policy recommendations are made.

## Keywords

Price discovery, volatility transmission, currency spot, currency futures, co-integration

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## Introduction

Price discovery mechanism explains in the short- and long-run prices in the spot and futures markets are systematically related (Hasbrouck, 1995; Kawaller, Koch & Koch, 1987; Stoll & Whaley, 1990b). This is so because assets traded in these markets are similar in nature. It is therefore not surprising that futures price is derived from the spot price of assets along with interest rate. Given this, any new information be it economic, political or company announcement can change prices either in the spot or in futures market. Following from this, existing studies (see Abhyankar, 1995; Antoniou & Holmes, 1995; Brooks et al., 2001; Chan, 1992; Ghosh, 1993; Min & Najand, 1999; Sehgal, Ahmed & Deisting, 2015) argue that future markets are the first to effected by new information and then spread to the spot market. Thus, futures price changes lead to changes in spot prices. Notwithstanding, it is quite possible that futures and spot prices change concurrently (see Bekiros & Diks, 2008; Somnath, 2011). Also, it is possible that neither of these two markets lead each other, in which case price discovery occurs equally in both markets. What is crucial, however, is that if one market process information faster than the other, a lead-lag relationship will exit (see Boyrie et al., 2012; Cabrera, Wang, & Yang, 2009; Chan & Gau, 2010; Chatrath & Song, 1998; Craln & Lee, 1995; Martens, Kofman, & Vorst, 1998; Rosenberg & Traub, 2006).

There are many reasons for immediate reaction of few markets on arrival of new information. Among the possible explanatory factors are mentioned: informational efficiency (efficiency of their functioning), higher liquidity, minimum transaction costs and the degree of integration between the two markets. For instance, if spot and futures markets are efficiently integrated, it is possible for volatility to be transmitted from one market to the other.

Across the globe, the issue of price discovery and volatility transmission between currency spot and futures markets has received high attention from academicians and policymakers (see Abhyankar, 1995; Antonakakis, 2012; Bessembinder & Seguin, 1992; Brooks, Rew & Ritson, 2001; Chan, 1992; Choudhary & Bajaj, 2013; Hasbrouck, 1995; Inagaki, 2007; Sehgal, Ahmed & Deisting, 2015; Stoll & Whaley, 1990a). Here it is interesting to see whether the futures markets reacts first to spot market or spot markets reacts over the future market to the same information. Several studies devoted to this issue have found mixed results. The flow depends on such factors as the trading platform, the nature and content of information flow, volatility, market size, trade volume, transparency, technology use, among others (see De Boyrie, Pavlova & Parhizgari, 2012). Many studies found that futures market lead spot market and that volatility transmission is from futures to spot market (Craln & Lee, 1995; De Boyrie et al., 2012; Martens, Kofman, & Vorst, 1998; Sehgal, Ahmed & Deisting, 2015). However, other studies have revealed that the spot price of exchange rate influences futures price (see Antoniou, Holmes, & Priestley, 1998; Cheung & Fung, 1997; Darrat, Rahman & Zhong, 2002; Herbst McCormack, & West, 1987; Ryoo & Smith, 2004; Shyy, Vijayraghavan & Scott-Quinn, 1996; Tan, Stevenson, Frino & Jarnecic, 2006).

In India, National Stock Exchange (NSE) was the first stock exchange, permitted by the Securities and Exchange Board of India (SEBI) to set up its separate currency derivatives segment. The NSE launched its currency futures trading of United States dollar/Indian Rupee (USD/INR) in August 2008. Similarly, the Bombay Stock Exchange (BSE) and Multi Commodity Exchange of India Ltd (MCX) introduced currency futures trading in October 2008. After a successful year of trading, the Reserve Bank of India (RBI) and SEBI allowed trade in currency futures in three new pairs, namely, Japanese Yen/Indian Rupee (JPY/INR), Great Britain Pound/Indian Rupee (GBP/INR) and Euro/Indian Rupee (EUR/INR). Since the inception, the currency futures market has experienced impressive growth in total turnover. The total turnover of NSE currency futures in 2008–2009 was about ₹1,622,724.3 million and the number of futures contracts stood 32,672,768 (see Table 1). In 2014–2015, the total turnover of currency futures has gone up to ₹22,479,923.4 million and that of contracts increased to 355,588,963.

**Table 1.** Total Turnover of Currency Futures and Options at NSE

| | Currency Futures | | Currency Options | | Total | |
|---|---|---|---|---|---|---|
| Year | No. of contracts | Turnover | No. of contracts | Notional Turnover | No. of contracts | Turnover |
| 2014–2015 | 355,588,963 | 2,247,992.34 | 125,075,731 | 775,915.32 | 480,664,694 | 3,023,907.67 |
| 2013–2014 | 478,301,579 | 2,940,885.92 | 181,890,951 | 1,071,627.54 | 660,192,530 | 4,012,513.45 |
| 2012–2013 | 684,159,263 | 3,765,105.33 | 275,084,185 | 1,509,359.32 | 959,243,448 | 5,274,464.65 |
| 2011–2012 | 701,371,974 | 3,378,488.92 | 271,972,158 | 1,296,500.98 | 973,344,132 | 4,674,989.91 |
| 2010–2011 | 712,181,928 | 3,279,002.13 | 37,420,147 | 170,785.59 | 749,602,075 | 3,449,787.72 |
| 2009–2010 | 378,606,983 | 1,782,608.04 | — | — | 378,606,983 | 1,782,608.04 |
| 2008–2009 | 32,672,768 | 162,272.43 | — | — | 32,672,768 | 162,272.43 |

**Source:** NSE website.

This outcome provides adequate grounds for a comprehensive study of price discovery in India's currency futures and spot markets.

The rest of the article is organized as follows. The next section is devoted to a brief survey of the theoretical and empirical literature. The third section provides the objective which is followed by the rationale of the study. The fifth section describes the data and the methodology. The analysis of the results is presented in the sixth section while in the last and final section, we offer concluding remarks with managerial implications.

## Review of Literature

Fragmentation which in itself comes with institutional, economic, regulatory issues has resulted in a situation where securities no longer trade in single centralized market (Cohen, Conroy & Maier, 1985; Cohen, Maier, Schwartz & Whitcomb, 1982; Harris, 1993; Mendelson, 1987; Shapiro, 1993). This has provided 'fertile grounds' for price discovery—a situation where information is transmitted from one market to the other which eventually determines the price of a security (see Hasbrouck, 1995). The existence of price discovery creates lead-lag relationships between futures and spots markets (Theissen, 2011). The resulting lead-lag relationship demonstrates how fast new information moves between markets as well as the level of linkages between them (Chan, 1992). There is however no lead-lag relationship in a perfectly frictionless world. As Chan (1992) puts it, 'in a perfectly competitive world, price movements of the two markets are contemporaneously correlated and not cross-autocorrelated' (Chan, 1992, p. 124). The lead-lag relationship is influenced by such factors as the time-lag of information flow between markets, the trading volume or intensity within the two markets, and the extent to which the information disseminates within the markets (i.e., either firm specific or market wide). Essentially, price discovery depends on the market in which information affects first. The speed and degree of influence of information flow in determining prices eventually explain market efficiency (see Mahalik, Acharya & Babu, 2009; Sudhakar & Srikanth, 2016; Zapata, Fortenbery & Armstrong, 2005; Zhong, Darrat & Otero, 2004). Moreover, the extent to which prices display market information clearly explains the rate at which such information is disseminated among participants in a market (Zapata et al., 2005). In summary, 'in a static sense, price discovery implies the existence of equilibrium price and in a dynamic sense, the price discovery process describes how information is produced and transmitted across the markets' (Mahalik et al., 2009, p. 4). The 'strength' of price discovery is an important source of information for hedgers and arbitrageurs (see Kenourgios & Samitas, 2004; Unlu & Ersoy, 2012).

Bearing these considerations in mind, empirical evidence is required to determine which of these markets leads the other. Several authors have therefore investigated price discovery and volatility spillover between currency spot and futures markets. Yiuman Tse, Xiang and Fung (2006) analysed the price discovery of Chicago Mercantile Exchange (CME) floor-traded futures, electronic Globex futures and electronic retail online spot trading in the Euro and Japanese yen markets by using Hasbrouck and Gonzalo and Granger information share models. High frequency data of spot and futures prices of US dollar-Euro and US Dollar-Japanese Yen are collected from May 2004 to July 2004. The study concludes that currency futures market plays an important role in price discovery process by incorporating information from online spot trading. Raghavendra and Velmurugan (2013) investigated the short-run relationship between currency spot and futures prices of GBP/INR by employing Granger causality test. The study used daily currency futures and spot prices of GBP/INR from February 2010 to December 2012 for the empirical analysis. The results of Granger causality test show that spot price of GBP/INR does not Granger cause futures prices, however, futures prices influences spot prices. The result concludes that there is a unidirectional causal relationship from futures market to spot market of GBP/INR currency pair traded in India. Unlu and Ersoy (2012) examined the short- and long-run relationship between currency spot and futures market in Turkey. The Engle and Granger (1987) co-integration test and the error correction model (ECM) were used for the estimation. Daily spot and futures prices of TRY-US dollar for the periods January 2007 to December 2011 was used. The results confirm that there is bidirectional long-run relationship between currency spot and futures market. Rosenberg and Traub (2006) used Hasbrouck and Gonzalo and Granger information share models to analysis price discovery in the currency spot and futures prices of Deutsche Mark, Japanese Yen, Swiss Franc and the Pound. Daily futures prices were collected from US CME in the time and sales reports for the periods May 1996 and August 1996. The study finds that currency futures market leads spot market and also plays an important role in price discovery process. Somnath (2011) investigates the relationship between currency futures trading activity and volatility in the Indian spot market by using the generalized autoregressive conditional heteroskedasticity (GARCH) framework. Daily spot and futures prices of USD/INR for the periods 29 August 2008 to 10 February 2010 were used for the analysis. The results reveal that there is bidirectional causality between the volatility in spot prices and the trading activity in the currency futures market. Chen and Gau (2008) examined the price discovery in the spot and futures market in USA. They used intraday data of spot and futures prices of euro/dollar and Japanese yen/dollar from 1 January 2004 to 31 December 2004. They concluded that spot market play dominant role in price discovery process than futures market. Sriram and Senthil (2013) explored price discovery and volatility spillover in spot and futures currency market by the use of co-integration, GARCH and ECMs. They used daily spot and futures prices of USD/INR from 1 June 2009 to 10 July 2013. Their result shows that there is unidirectional causality from spot to futures market. They concluded that volatility transmission move from the spot market to the futures market; whereas there was no evidence of volatility spillover from futures to spot market.

## Objective of the study

This study investigates the short- and long-run association between India's currency spot and futures markets as well as the volatility transmission between these markets. The Engle–Granger co-integration test, ECM, Granger causality test and the bivariate GARCH were used for the estimation.

## Rationale of the Study

It is important to note that although the currency derivative market in some developing countries and mainly emerging markets have been growing steadily in recent years, the literature relating to price discovery and volatility transmission in currency spot and futures markets has mainly been confined to developed economies. Although some studies on the same topic has been conducted in India and other emerging markets, the present study is devoted to this topic for fairly large period in India. It covers the aftermath of global financial crisis and European debt crisis to bridge the important gap in the literature in a comprehensive way.

## Data and Methodology

### *Data*

The study used daily data from 1 March 2010 to 20 January 2015 to investigate price discovery and volatility spillover in currency spot and futures market in India. Daily futures prices of USD-INR, Yen-INR, Pound-INR and Euro-INR were sourced from the official website of NSE of India. The spot price of USD-INR, Yen-INR, Pound-INR and Euro-INR are obtained from the RBI. The study considers only near-month futures prices since most trading activities take place in this contract. All closing prices of spot and futures are expressed in natural logarithms.

### *Methodology*

To investigate the long- and short-run relationship between currency spot and futures prices of USD/INR, JPY/INR, GBP/INR and EUR/INR and volatility spillover between these prices, the study used Engle and Granger (1987) co-integration analysis, ECM, Granger causality test and the bivariate GARCH model.

Generally speaking, the first step in co-integration analysis is to verify the order of integration of each price by the use of unit root test. To do this, the augmented Dickey–Fuller (ADF) (1981) and Phillips and Perron (PP) (1988) unit root tests are used. If the two series are of the same order of integration, then an appropriate co-integration approach is used to test for the possibility of a co-integrating relationship. In our case, the Engle and Granger (1987) co-integration approach which explains the long-run relationship between two variables is used. Similar to all co-integration procedures, the first step in Engle and Granger (1987) co-integration analysis is to determine the order of integration of each price. In the second step, co-integrating equation is identified and estimated using OLS. In the third and final step, the residuals from OLS regression are obtained to see whether they are stationary or not.

Once co-integration is confirmed, we proceed to apply an ECM that includes previous period error as well as lagged values of first differences of each variable. Consider the ECM specification, which can be written as follows:

$$\Delta F_t = \sum_{i=1}^{p-1} \alpha_{F,i} \Delta S_{t-i} + \sum_{i=1}^{p-1} \beta_{F,i} \Delta F_{t-i} + \theta_F z_{t-i} + \varepsilon_{F,t} \quad (1)$$

$$\Delta S_t = \sum_{i=1}^{p-1} \alpha_{S,i} \Delta S_{t-i} + \sum_{i=1}^{p-1} \beta_{S,i} \Delta F_{t-i} + \theta_S z_{t-i} + \varepsilon_{S,t} \quad (2)$$

For each of the two preceding equations, $\hat{z}_{t-1}$ is interpreted as the equilibrium error term. The coefficients $\theta_f$ and $\theta_s$ are the speed of adjustment coefficients which have important implications for the ECM. At least one speed adjustment coefficient must be non-zero for the model to be stable. For instance, if $\theta_f$ is statistically insignificant the current period changes in futures price does not last period's deviation from long-run equilibrium.

As earlier indicated the study intends to investigate the volatility spillover between currency spot and futures prices by the use of bivariate GARCH model. Volatility transmission is important in the study of information transmission because it is also a source of information to which investors react and form new expectations regarding risk and return. Two research papers (French & Roll, 1986; Ross, 1989) showed that variance is an important source of information which is directly related to the flow of information. The bivariate GARCH (1, 1) model can be written as follows.

$$H_t = \begin{pmatrix} h_{11,t} \\ h_{12,t} \\ h_{22,t} \end{pmatrix} + \begin{pmatrix} c_{11} \\ c_{12} \\ c_{22} \end{pmatrix} + \begin{pmatrix} a_{11} & 00 \\ 0 & a_{12}0 \\ 0 & 0a_{22} \end{pmatrix} + \begin{pmatrix} \varepsilon^1_{1,t-1} \\ \varepsilon_{1,t-1}\,\varepsilon_{2,1-1} \\ \varepsilon^2_{2,t-1} \end{pmatrix} + \begin{pmatrix} \beta_{11} & 00 \\ 0 & \beta_{12}0 \\ 0 & 0\beta_{22} \end{pmatrix} + \begin{pmatrix} h_{11,t-1} \\ h_{12,t-1} \\ h_{22,t-1} \end{pmatrix} \tag{3}$$

where the residual terms are distributed as $N(0, H_t)$. In a diagonal representation, the 2 × 2 conditional variance-covariance matrix $H_t$ under a GARCH (1, 1) specification is given.

The study follows a two-step procedure to examine the volatility transmission between currency spot and futures prices. The first step involves estimation of ECM and in the second step bivariate GARCH model. The expansion of the matrix $H_t$ is structured as:

$$h_{11t} = c_{11} + a_{11}\varepsilon^2_{1t-1} + \beta_{11}h_{t-1} + \theta_{11}\varepsilon^1_{1F,t-1} \tag{4}$$

$$h_{22t} = c_{22} + a_{22}\varepsilon^2_{1t-1} + \beta_{22}h_{t-1} + \theta_{22}\varepsilon^2_{2S,t-1} \tag{5}$$

$$h_{12t} = c_{12} + a_{12}\varepsilon^1_{1t-1}, \varepsilon^2_{2t-1} + \beta_{12}h_{t-1} + \theta_{12}h_{12,t-1} \tag{6}$$

In Equation 4, conditional variance for currency futures market includes the own lagged value of residuals along with past conditional variance and past lagged value of residuals from ECM. In this equation, $\theta_{11}$ is the coefficient representing the spillover from futures to spot market. In Equation 5 the conditional variance equation for spot market, $\theta_{22}$ is the coefficient representing the spillovers from spot to futures market. The Equation 6 covariance ($h_{12,t}$) for two markets returns to influence each other and coefficient $\theta_{12}$ representing volatility spillover between two markets.

## Empirical Analysis

Table 2 provides summary statistics, such as the sample mean, standard deviation, skewness, kurtosis and the Jarque–Bera test and LM test for the selected currency spot and futures returns of USD/INR, JPY/INR, GBP/INR and EUR/INR. The highest mean returns are observed in case of currency spot of USD/INR and EUR/INR and lowest currency spot returns for GBP/INR and futures returns of JPY/INR. The standard deviation of currency spot returns of USD/INR and EUR/INR is higher among all the four currencies. This implies that higher volatility is observed in these two markets compared to the others. The negative skewness coefficient for most of the currencies implies that the frequency distribution of the return series is negatively skewed. The kurtosis value exceeds four for all currencies, which shows that the distribution of returns is leptokurtic. The Jarque–Bera test statistics are statistically significant and hence we reject the null hypothesis of a normal distribution of the return series.

**Table 2.** Descriptive statistics

| Variables | Mean | Standard Deviation | Skewness | Kurtosis | JB Test | LM Test |
|---|---|---|---|---|---|---|
| USD/INR (Spot) | 0.00328 | 0.85431 | −0.9721 | 4.2901 | 228.19 | 58.312 (0.000) |
| USD/INR (Futures) | 0.000214 | 0.6210 | −2.4213 | 12.4913 | 1456.30 | 47.843 (0.000) |
| JPY/INR (Spot) | −0.00432 | 0.0712 | −0.0219 | 6.4140 | 583.32 | 23.781 (0.001) |
| JPY/INR (Futures) | 0.7321 | 0.0612 | 2.31902 | 18.3104 | 856.00 | 82.732 (0.000) |
| GBP/INR (Spot) | −0.5432 | 0.00321 | −3.9010 | 8.3105 | 3567.32 | 66.829 (0.000) |
| GBP/INR (Futures) | 0.00432 | 0.03419 | −0.8213 | 15.4505 | 238.02 | 21.632 (0.001) |
| EUR/INR (Spot) | 0.9531 | 0.9921 | −4.5404 | 22.3493 | 383.210 | 34.781 (0.000) |
| EUR/INR (Futures) | 0.0712 | 0.2154 | −3.5204 | 7.59012 | 32.219 | 29.283 (0.000) |

**Source:** Authors' calculations.

To determine the order of integration of each variable, the ADF and PP unit root tests were employed. The results of the ADF and PP unit root tests given in Table 3 show that log series of currency spot and futures of USD/INR, JPY/INR, GBP/INR and EUR/INR are non-stationary at levels but they become stationary at first difference ($s = −32.910, $f = 28.452, ¥s = −35. 341, ¥f = −46.4190, £s = 26.430 and £f = −53.671). Thus, all the series are integrated of the first order (i.e., $I(1)$). This implies that these series have possible long-run relationship to be examined by the co-integration test.

The Engle and Granger (1987) co-integration test is applied to check whether currency spot and futures price of USD/INR, JPY/INR, GBP/INR and EUR/INR are cointegrated or not. The results of Engle and Granger (1987) co-integration test are given in Table 4. It reveals that the residual values of USD/INR, JPY/INR, GBP/INR and EUR/INR are stationary at 1 per cent level. This implies that long-run relationship exist between currency spot and futures prices of USD/INR, JPY/INR, GBP/INR and EUR/INR. This result shows that both foreign currency spot and futures prices are cointegrated and move together in the long run. The implication of co-integration is that the currencies in these two separate markets respond disproportionately to the pricing information in the short run, but they converge to equilibrium in the long run under the condition that both markets are innovative and efficient.

The equilibrium relationship between the non-stationary variables is used to estimate the ECM. If two variables (prices) deviate from short run equilibrium level it may lead to arbitrage opportunities in Indian currency market. These arbitrage opportunities help in correcting the disequilibrium between currency spot and futures prices of USD/INR, JPY/INR, GBP/INR and EUR/INR. Table 5 shows that the estimates of the speed of adjustment coefficients between two variables are identified by ECM estimation using Equations (1) and (2). The result shows that the coefficients of the error correction terms are higher in the case of currency spot of JPY/INR, GBP/INR and EURO/INR and are statistically significant ($\theta_{syen} = 0.8723$, $\theta_{spound} = 0.3809$ and $\theta_{seuro} = 0.2292$) when compared to the futures price. It reveals that currency spot price corrects more disequilibrium between the spot and the futures prices. Thus, spot market is adjusting to the new information faster than futures market suggesting that currency spot price leads the futures price and contributes largely to price discovery. Finally, the results show that coefficients of

**Table 3.** Results of unit root tests

| Variables | ADF in Level | ADF in First Difference | PP in Level | PP in First Difference |
|---|---|---|---|---|
| USD/INR (Spot) | −2.691 | −32.910* | −2.341 | −33.512* |
| | (0.1490) | (0.000) | (0.4231) | (0.0001) |
| USD/INR (Futures) | −1.8410 | −28.452* | −0.710 | −28.319* |
| | (0.3679) | (0.000) | (0.367) | (0.0001) |
| JPY/INR (Spot) | −1.632 | −35.341* | −1.526 | −35.431* |
| | (0.2831) | (0.000) | (0.521) | (0.0001) |
| JPY/INR (Futures) | −1.3750 | −46.4190* | −1.143 | −47.410* |
| | (0.4390) | (0.000) | (0.3410) | (0.0001) |
| GBP/INR (Spot) | −0.2134 | −26.430* | −1.6880 | −26.721* |
| | (0.5210) | (0.000) | (0.213) | (0.000) |
| GBP/INR (Futures) | −1.319 | −53.671* | −0.8120 | −54.321* |
| | (0.4210) | (0.000) | (0.314) | (−0.000) |
| EUR/INR (Spot) | −1.4330 | 44.710* | −2.901 | −43.320* |
| | (0.3190) | (0.000) | (0.1810) | (0.000) |
| EUR/INR (Futures) | −0.5230 | 71.310* | −1.3240 | −71.848* |
| | (0.6124) | (0.000) | (0.231) | (0.000) |

**Source:** Authors' calculations.
**Note:** * indicate 1% level of significance.

**Table 4.** Results of Engle–Granger Co-integration Test

| Co-integration Regression between Spot and Futures prices of USD-INR ($S_t = \beta_o + \beta_1 F_t + u_t$) | | | |
|---|---|---|---|
| Variables | Coefficient | *t*-statistic | *p*-value |
| Constant | 0.0183 | 1.8120 | 0.0955 |
| $\beta_1$ | 2.610 | 9.1290 | 0.0000 |
| $R^2 = 0.889$ | DW statistic = 0.9218 | | |
| Residual based on co-integration test ($\Delta\hat{u} = \beta\hat{u}_{t-1} + v_t$) | | | |
| Variable | ADF-stat | *p*-value | |
| $\hat{u}$ | −26.3960* | 0.0000 | |
| Co-integration regression between spot and futures prices of Japanese Yen-INR ($S_t = \beta_o + \beta_1 F_t + u_t$) | | | |
| Variables | Coefficient | *t*-statistic | *p*-value |
| Constant | 0.3210 | 2.519 | 0.0052 |
| $\beta_1$ | 1.3604 | 7.321 | 0.0000 |
| $R^2 = 0.790$ | DW statistic = 0.9210 | | |
| Residual based on co-integration test ($\Delta\hat{u} = \beta\hat{u}_{t-1} + v_t$) | | | |
| Variable | ADF-stat | *p*-value | |
| $\hat{u}$ | −19.5280* | 0.0000 | |
| Co-integration regression between Spot and futures prices of pound sterling-INR ($S_t = \beta_o + \beta_1 F_t + u_t$) | | | |
| Variables | Coefficient | *t*-statistic | *p*-value |
| Constant | −0.8430 | −3.7201 | 0.0000 |
| $\beta_1$ | 2.4620 | 8.530 | 0.0000 |
| $R^2 = 0.8190$ | DW statistic | 0.7610 | |

| Residual based on co-integration test ($\Delta\hat{u} = \beta\hat{u}_{t-1} + v_t$) | | | |
|---|---|---|---|
| Variable | ADF-statistic | *p*-value | |
| $\hat{u}$ | −41.3401* | 0.000 | |
| Co-integration regression between spot and futures prices of Euro-INR ($S_t = \beta_o + \beta_1 F_t + u_t$) | | | |
| Variables | Coefficient | *t*-statistic | *p*-value |
| Constant | 0.00394 | 1.4587 | 0.3180 |
| $\beta_1$ | 4.6570 | 10.569 | 0.0000 |
| $R^2 = 0.920$ | DW statistic | 0.8651 | |
| Residual based on co-integration test ($\Delta\hat{u} = \beta\hat{u}_{t-1} + v_t$) | | | |
| Variable | ADF-statistic | *p*-value | |
| $\hat{u}$ | −37.9210* | 0.000 | |

**Source:** Authors' calculations.
**Note:** * indicates 1% level of significance.

**Table 5.** Results of Error Correction Model

| | USD/INR | | | | JPY/INR | | | |
|---|---|---|---|---|---|---|---|---|
| | $\Delta S_t$ | | $\Delta F_t$ | | $\Delta S_t$ | | $\Delta F_t$ | |
| | Coefficients | | Coefficients | | Coefficients | | Coefficients | |
| $Z_{t-1}$ | $\theta_s$ | 0.812* (5.450) | $\theta_f$ | 0.565* (2.689) | $\theta_s$ | 0.872* (3.276) | $\theta_f$ | 0.0004 (1.110) |
| $\Delta S_{t-1}$ | $\alpha_{s,1}$ | 2.992 (7.89) | $\alpha_{f,1}$ | 0.412 (6.734) | $\alpha_{s,1}$ | 0.890 (7.481) | $\alpha_{f,1}$ | 0.004 (1.328) |
| $\Delta S_{t-2}$ | $\alpha_{s,2}$ | −0.026 (−1.578) | $\alpha_{f,2}$ | 0.0012 (0.737) | $\alpha_{s,2}$ | −0.0005 (−0.973) | $\alpha_{f,2}$ | −0.002 (−1.378) |
| $\Delta F_{t-1}$ | $\beta_{s,1}$ | 0.4810 (9.390) | $\beta_{f,1}$ | 0.076 (1.621) | $\beta_{s,1}$ | 0.301 (2.439) | $\beta_{f,1}$ | 0.005 (1.638) |
| $\Delta F_{t-2}$ | $\beta_{s,1}$ | −0.0390 (−0.367) | $\beta_{f,2}$ | 0.003 (1.566) | $\beta_{s,1}$ | 0.004 (0.793) | $\beta_{f,2}$ | 0.007 (0.839) |
| | GBP/INR | | | | EUR/INR | | | |
| | $\Delta S_t$ | | $\Delta F_t$ | | $\Delta S_t$ | | $\Delta F_t$ | |
| | Coefficients | | Coefficients | | Coefficients | | Coefficients | |
| $Z_{t-1}$ | $\theta_s$ | 0.381* (2.279) | $\theta_f$ | 0.002 (2.459) | $\theta_s$ | 0.229* (3.748) | $\theta_f$ | 0.004 (1.380) |
| $\Delta S_{t-1}$ | $\alpha_{s,1}$ | 3.728 (9.170) | $\alpha_{f,1}$ | 0.412 (6.734) | $\alpha_{s,1}$ | 0.890 (7.481) | $\alpha_{f,1}$ | 0.005 (1.845) |
| $\Delta S_{t-2}$ | $\alpha_{s,2}$ | 0.0002 (0.983) | $\alpha_{f,2}$ | −0.007 (0.3870) | $\alpha_{s,2}$ | −0.0005 (−0.973) | $\alpha_{f,2}$ | 0.034 (2.960) |
| $\Delta F_{t-1}$ | $\beta_{s,1}$ | 0.047 (2.259) | $\beta_{f,1}$ | 0.0013 (0.9827) | $\beta_{s,1}$ | 0.0009 (0.439) | $\beta_{f,1}$ | 0.028 (2.990) |
| $\Delta F_{t-2}$ | $\beta_{s,1}$ | −0.0002 (−0.218) | $\beta_{f,2}$ | −0.0001 (1.231) | $\beta_{s,1}$ | 0.078 (1.276) | $\beta_{f,2}$ | −0.099 (−2.780) |

**Source:** Authors' calculations.
**Notes:** $Z_{t-1}$ represents the lagged error correction term. * indicates 1% level of significance. Figures in parentheses are *t*-values. Lag length of 2 is chosen based on Akaike and Schwarz information criterion (AIC) and final predication error (FPE).

the error correction term between spot and futures prices of USD/INR are significant in both equations ($\theta_{sUSD} = 0.8120$ and $\theta_{fUSD} = 0.5654$) suggesting a bidirectional error correction. In this case also, the coefficient of the error correction term is higher in the case of currency spot of USD/INR when compared with futures prices of USD/INR.

The Granger causality test was used to investigate the causal relationship between foreign currency spot and futures prices of USD/INR, JPY/INR, GBP/INR and EUR/INR. The results of Granger causality test presented in Table 6 reveals a bidirectional causal relationship between currency spot and futures prices of USD/INR. However, there is a unidirectional causal relationship from currency spot to futures prices of JPY/INR, GBP/INR and EUR/INR. Thus, currency spot prices Granger causes futures prices of USD/INR, JPY/INR, GBP/INR and EUR/INR.

The results of the bivariate GARCH model are presented in Tables 7–10. It shows that there exists a significant unidirectional volatility spillover from currency spot to futures returns of JPY/INR,

**Table 6.** Results of Granger Causality Test

| Hypothesis | *F*-statistic | *P*-value |
|---|---|---|
| Spot does not Granger Cause Futures prices of USD/INR | 13.391 | 0.000 |
| USD/INR Futures does not Granger cause spot price | 8.464 | 0.000 |
| Spot does not Granger cause futures prices of JPY/INR | 17.483 | 0.000 |
| JPY/INR Futures does not Granger Cause spot price | 0.034 | 0.382 |
| Spot does not Granger cause Futures prices of GBP/INR | 21.769 | 0.000 |
| Futures price of GBP/INR does not Granger cause spot prices | 0.279 | 0.181 |
| Spot does not Granger cause futures EUR/INR | 9.730 | 0.000 |
| Futures prices of EUR/INR does not Granger cause spot prices | 0.009 | 0.413 |

**Source:** Authors' calculations.

**Note:** Lag length of 2 is chosen based on Akaike and Schwarz information criterion (AIC) and final predication error (FPE).

**Table 7.** Results of Bivariate GARCH Model between Spot and Futures Price of USD-INR

| Variable | Coefficients | *Z*-Value | *P*-value |
|---|---|---|---|
| $\theta_{11}$ | 0.372 | 2.832 | 0.000 |
| $\theta_{21}$ | 2.327 | 3.890 | 0.000 |
| $\theta_{12}$ | 0.753 | 2.784 | 0.000 |
| $\theta_{22}$ | 0.265 | 1.832 | 0.034 |
| $\alpha_{11}$ | 0.321 | 4.543 | 0.000 |
| $\alpha_{22}$ | 0.998 | 6.832 | 0.000 |
| $\beta_{11}$ | −0.218 | −3.333 | 0.000 |
| $\beta_{22}$ | 0.007 | 2.812 | 0.000 |
| $\theta_{11(futures)}$ | 0.067 | 2.121* | 0.000 |
| $\theta_{22\ (spot)}$ | 0.871 | 2.456* | 0.000 |
| Residual Diagnostics | Test Statistic | | *P*-value |
| LBQ (12) | 12.452 | | 0.034 |
| $LBQ^2$ (12) | 18.439 | | 0.077 |
| LM test | 3.231 | | 0.279 |

**Source:** Authors' calculations.

**Notes:** LM is the ARCH effect in the standardized squared residuals up to six lags. Estimates of constant and covariance terms are not reported in the table.
* implies 1% level of significance.

**Table 8.** Results of Bivariate GARCH Model between Spot and Futures Price of JPY/INR

| Variable | Coefficients | *Z*-value | *P*-value |
|---|---|---|---|
| $\theta_{11}$ | 0.004 | 1.436 | 0.041 |
| $\theta_{21}$ | −0.843 | −2.439 | 0.000 |
| $\theta_{12}$ | 0.007 | 1.548 | 0.027 |
| $\theta_{22}$ | 0.544 | 4.543 | 0.000 |
| $\alpha_{11}$ | −0.457 | −3.655 | 0.000 |
| $\alpha_{22}$ | 0.007 | 2.765 | 0.000 |
| $\beta_{11}$ | 0.732 | 6.984 | 0.000 |
| $\beta_{22}$ | 0.021 | 4.891 | 0.000 |
| $\theta_{11(\text{futures})}$ | −0.0004 | −0.113 | 0.532 |
| $\theta_{22\ (\text{spot})}$ | 0.027 | 3.654* | 0.000 |
| Residual Diagnostics | Test Statistic | | *P*-value |
| LBQ (12) | 9.899 | | 0.218 |
| $LBQ^2$ (12) | 21.329 | | 0.067 |
| LM test | 4.892 | | 0.892 |

**Source:** Authors' calculations.

**Notes:** LM is the ARCH effect in the standardized squared residuals up to 6 lags. Estimates of constant and covariance terms are not reported in the table. * implies 1% level of significance.

**Table 9.** Results of Bivariate GARCH Model between Spot and Futures Price of GBP/INR

| Variable | Coefficients | *Z*-value | *P*-value |
|---|---|---|---|
| $\theta_{11}$ | 0.0002 | 1.147 | 0.244 |
| $\theta_{21}$ | 0.094 | 3.732 | 0.000 |
| $\theta_{12}$ | 0.472 | 6.644 | 0.000 |
| $\theta_{22}$ | −0.433 | −2.658 | 0.000 |
| $\alpha_{11}$ | 0.008 | 9.543 | 0.000 |
| $\alpha_{22}$ | 0.033 | 4.854 | 0.000 |
| $\beta_{11}$ | 0.0004 | 0.843 | 0.328 |
| $\beta_{22}$ | 0.153 | 7.218 | 0.000 |
| $\theta_{11(\text{futures})}$ | −0.002 | −0.106 | 0.644 |
| $\theta_{22\ (\text{spot})}$ | 0.633 | 4.902* | 0.000 |
| Residual Diagnostics | Test Statistic | | *P*-value |
| LBQ (12) | 7.432 | | 0.376 |
| $LBQ^2$ (12) | 11.832 | | 0.197 |
| LM test | 6.322 | | 0.321 |

**Source:** Authors' calculations.

**Notes:** LM is the ARCH effect in the standardized squared residuals up to six lags. Estimates of constant and covariance terms are not reported in the table. * implies 1% level of significance.

**Table 10.** Results of Bivariate GARCH Model between Spot and Futures Price of EUR/INR

| Variable | Coefficients | *Z*-value | *P*-value |
|---|---|---|---|
| $\theta_{11}$ | 0.027 | 3.443 | 0.000 |
| $\theta_{21}$ | 0.004 | 2.327 | 0.000 |
| $\theta_{12}$ | −0.921 | −5.432 | 0.000 |
| $\theta_{22}$ | 0.071 | 2.499 | 0.000 |
| $\alpha_{11}$ | 0.328 | 7.833 | 0.000 |
| $\alpha_{22}$ | 0.833 | 4.009 | 0.000 |
| $\beta_{11}$ | 0.0096 | 6.829 | 0.000 |
| $\beta_{22}$ | 0.328 | 5.867 | 0.000 |
| $\theta_{11(\text{futures})}$ | 0.007 | 1.379 | 0.532 |
| $\theta_{22\ (\text{spot})}$ | 0.067 | 3.731* | 0.000 |
| Residual Diagnostics | Test Statistic | | *P*-value |
| LBQ (12) | 4.754 | | 0.332 |
| $LBQ^2$ (12) | 9.321 | | 0.257 |
| LM test | 13.453 | | 0.293 |

**Source:** Authors' calculations.

**Notes:** LM is the ARCH effect in the standardized squared residuals up to 6 lags. Estimates of constant and covariance terms are not reported in the table. * implies 1% level of significance.

GBP/INR and EUR/INR ($\theta_{22\text{yen}} = 0.027$, $\theta_{22\ \text{pond}} = 0.633$ and $\theta_{22\text{Euro}} = 0.067$. On the other hand, no reverse volatility spillover is found from futures returns to currency spot returns of JPY/INR, GBP/INR and EUR/INR since the *Z* values are not statistically significant. This is due to the fact that new information is first reflected in currency spot market and then transmitted to futures market. However, bidirectional volatility transmission is found between currency spot and futures returns of USD/INR ($\theta_{22s} = 0.8710$, $\theta_{22f} = 0.0971$). The positive spillover coefficients ($\theta_{11}$ and $\theta_{22}$) show that news to one market increase the conditional variance of another in the next period. Finally, result suggests the spillover coefficient from currency spot to futures returns is greater than the spillover coefficient from futures to currency spot prices.

## Conclusion

This article has examined price discovery and volatility transmission between currency spot and futures prices in India. Daily closing currency spot and futures prices of USD/INR, JPY/INR, GBP/INR and EUR/INR for the period 1 March 2010 to 20 January 2015 was used for the empirical analysis. Both the long- and short-run relationships as well as volatility transmission were investigated by the use of Engle–Granger co-integration test, ECM, Granger causality test and the bivariate GARCH model. The result of Engle–Granger co-integration test reveals there is a long-run relationship between currency spot and futures prices. The results of Granger causality test show that there exists unidirectional causal relationship from currency spot to futures prices of JPY/INR, GBP/INR and EUR/INR and bidirectional causal relationship between USD/INR currency spot and futures prices. The results further show that spot market is adjusting to the new information faster than futures market suggesting that spot price leads the

futures price and contributes largely to price discovery. Finally, the results show that that there is unidirectional volatility transmission from currency spot to futures prices of JPY/INR, GBP/INR and EUR/INR and bidirectional spillover between currency spot and futures prices of USD/INR.

## Managerial Implications

The study concludes that Indian currency futures market is inefficient as prices of currencies fails to reflect all the available information. This may be due to reason that speculators actively participating in the currency futures market leading to failure of efficient market hypothesis. In this context, the policy-makers are need to be ensured the stability in the markets to achieve efficient price discovery process. In addition, futures markets need to reflect all the available information while fixing up the prices of futures.

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### References

Abhyankar, A.H. (1995). Return and volatility dynamics in the FTSE 100 stock index and stock index futures markets. *Journal of Futures Markets, 15*(4), 457–488.

Antonakakis, N. (2012). Exchange return co-movements and volatility spillovers before and after the introduction of euro. *Journal of International Financial Markets, Institutions and Money, 22*(5), 1091–1109.

Antoniou, A., & Holmes, P. (1995). Futures trading, information and spot price volatility: Evidence for the FTSE-100 stock index futures contract using GARCH. *Journal of Banking and Finance, 19*(1), 117–129.

Antoniou, A., Holmes, P., & Priestley, R. (1998). The effects of stock index futures trading on stock index volatility: An analysis of the asymmetric response of volatility to news. *Journal of Futures Markets, 18*(2), 151–166.

Bekiros, S.D., & Diks, C.G. (2008). The relationship between crude oil spot and futures prices: Cointegration, linear and nonlinear causality. *Energy Economics, 30*(5), 2673–2685.

Bessembinder, H., & Seguin, P.J. (1992). Futures-trading activity and stock price volatility. *The Journal of Finance, 47*(5), 2015–2034.

Brooks, C., Rew, A.G., & Ritson, S. (2001). A trading strategy based on the lead–lag relationship between the spot index and futures contract for the FTSE 100. *International Journal of Forecasting, 17*(1), 31–44.

Cabrera, J., Wang, T., & Yang, J. (2009). Do futures lead price discovery in electronic foreign exchange markets? *Journal of Futures Markets, 29*(2), 137–156.

Chan, K. (1992). A further analysis of the lead–lag relationship between the cash market and stock index futures market. *The Review of Financial Studies, 5*(1), 123–152.

Chen, Y.L., & Gau, Y.F. (2008). *Price discovery in currency spot and futures markets* (Working Paper). Taiwan: National Central University.

Cheung, Y.W., & Fung, H.G. (1997). Information flows between Eurodollar spot and futures markets. *Multinational Finance Journal, 1*(4), 255–271.

Choudhary, K., & Bajaj, S. (2013). Price discovery process in nifty spot and futures markets. *Global Business Review, 14*(1), 55–88.

Cohen, K.J., Conroy, R.M., & Maier, S.F. (1985). Order flow and the quality of the market. In Y. Amihud, T.S.Y. Ho, & R.A. Schwartz (Eds), *Market making and the changing structure of the securities industry* (pp. 93–112). Lexington, MA: Lexington Books.

Cohen, K.J., Maier, S.F., Schwartz, R.A., & Whitcomb, D.K. (1982). An analysis of the economic justification for consolidation in a secondary security market. *Journal of Banking & Finance*, *6*(1), 117–136.

Craln, S.J., & Lee, J.H. (1995). Intraday volatility in interest rate and foreign exchange spot and futures markets. *Journal of Futures Markets, 15*(4), 395–421.

Darrat, A.F., Rahman, S., & Zhong, M. (2002). On the role of futures trading in spot market fluctuations: Perpetrator of volatility or victim of regret? *Journal of Financial Research, 25*(3), 431–444.

De Boyrie, M.E., Pavlova, I., & Parhizgari, A.M. (2012). Price discovery in currency markets: Evidence from three emerging markets. *International Journal of Economics and Finance, 4*(12), 61–75.

Engle, R.F., & Granger, C.W.J. (1997). Co-integration and error correction: Representation, estimation, and testing. *Econometrica, 55*(2), 251–276.

French, K.R., & Roll, R. (1986). Stock return variances: the arrival of information and reaction of traders. *Journal of Financial Economics, 17*(1), 5–26.

Ghosh, A. (1993). Hedging with stock index futures: Estimation and forecasting with error correction model. *Journal of Futures Market, 13*(7), 743–752.

Harris, L.E. (1993). Consolidation, fragmentation, segmentation, and regulation. *Financial Markets, Institutions and Instruments, 2*(5), 1–28.

Hasbrouck, J. (1995). One security, many markets: Determining the contributions to price discovery. *Journal of Finance, 50*(4), 1175–1199.

Herbst, A.F., McCormack, J.P., & West, E.N. (1987). Investigation of a lead–lag relationship between spot stock indices and their futures contracts. *Journal of Futures Markets, 7*(4), 373–381.

Inagaki, K. (2007). Testing for volatility spillover between the British pound and the euro. *Research in International Business and Finance*, *21*(2), 161–174.

Kawaller, I.G., Koch, P.D., & Koch, T.W. (1987). The temporal price relationship between S&P 500 futures and the S&P 500 index. *The Journal of Finance, 42*(5), 1309–1329.

Kenourgios, D., & Samitas, A. (2004). Testing efficiency of the copper futures market: New evidence from London metal exchange. *Global Business and Economics Review*, 261–271. Retrieved from http://users.uoa.gr/~dkenourg/proofs_Global_Bus.Econ.Review.PDF

Mahalik, M.K., Acharya, D., & Babu, M.S. (2009). *Price discovery and volatility spillovers in futures and spot commodity markets: Some empirical evidence from India* (IGIDR Proceedings/Project Reports Series No. PP-062-10). Mumbai: IGIDR. Retrieved 25 May 2016, from http://www.igidr.ac.in/pdf/publication/PP-062-10.pdf

Martens, M., Kofman, P., & Vorst, T.C. (1998). A threshold error-correction model for intraday futures and index returns. *Journal of Applied Econometrics, 13*(3), 245–263.

Mendelson, H. (1987). Consolidation, fragmentation, and market performance. *Journal of Financial and Quantitative Analysis, 22*(2), 189–207.

Min, J.H., & Najand, M. (1999). A further investigation of the lead–lag relationship between the spot market and stock index futures: Early evidence from Korea. *Journal of Futures Markets, 19*(2), 217–232.

Raghavendra, R.H., & Velmurugan, P.S. (2013). A study on causal relationship between spot return and future return of GBP/INR currency pair traded in India. *International Journal of Research in Business Management, 1*(6), 43–48.

Rosenberg, J.V., & Traub, L.G. (2006). *Price discovery in the foreign currency futures and spot market* (FRB of New York Staff Report No. 26). New York: Federal Reserve Bank of New York.

Ross, S. (1989). Information and volatility: the no-arbitrage martingale approach to timing and resolution irrelevancy. *Journal of Finance*, *44*(1), 1–17.

Ryoo, H.J., & Smith, G. (2004). The impact of stock index futures on the Korean stock market. *Applied Financial Economics, 14*(4), 243–251.

Sehgal, S., Ahmed, W., & Deisting, F. (2015). An investigation of price discovery and volatility spillovers in India's foreign exchange market. *Journal of Economic Studies, 42*(2), 261–284.

Shapiro, J. (1993). *Recent competitive developments in U.S. equity markets* (New York Stock Exchange Working Paper No. 93-02). New York: New York Stock Exchange.

Shyy, G., Vijayraghavan, V., & Scott-Quinn, B. (1996). A further investigation of the lead–lag relationship between the cash market and stock index futures market with the use of bid/ask quotes: The case of France. *Journal of Futures Markets, 16*(4), 405–420.

Somnath, S. (2011). *An empirical analysis of the relationship between currency futures and exchange rates volatility in India* (RBI Working Paper Series, WPS DEAP, 1/2011). Mumbai: RBI.

Sriram, M., & Senthil, M. (2013). Market efficiency and volatility spillover in spot and futures currency market. *International Journal of Management Excellence, 1*(3), 45–53.

Stoll, H.R., & Whaley, R.E. (1990a). The dynamics of stock index and stock index futures returns. *Journal of Financial and Quantitative Analysis, 25*(4), 441–468.

———. (1990b). Stock market structure and volatility. *Review of Financial Studies*, *3*(1), 37–71.

Sudhakar, A., & Srikanth, P. (2016). Determinants of in-the-money expiration of call option contracts—An empirical evidence from call options on nifty 50 index. *Global Business Review, 17*(6), 1373–1387.

Tan, A., Stevenson, M., Frino, A., & Jarnecic, E. (2006). Sources of price discovery in the Australian dollar currency market. *Review of Futures Markets, 15*(1), 45–83.

Theissen, E. (2011). Price discovery in spot and futures markets: A reconsideration. *The European Journal of Finance, 18*(10), 1–19.

Tse, Y., Xiang, J., & Fung, J.K. (2006). Price discovery in the foreign exchange futures market. *Journal of Futures Markets, 26*(11), 1131–1143.

Unlu, U. & Ersoy, E. (2012). Causal relationship between foreign currency futures and spot markets from Turkey. *Investment Management and Financial Innovations, 9*(2), 208–213.

Zapata, H., Fortenbery, T.R., & Armstrong, D. (2005, March). *Price discovery in the world sugar futures and cash markets: Implications for Dominican Republic* (Staff Working Paper No. 469). Madison, Wisconsin: Department of Agricultural & Applied Economics, University of Wisconsin-Madison.

Zhong, M., Darrat, A.F., & Otero, R. (2004). Price discovery and volatility spillovers in index futures markets: Some evidence from Mexico. *Journal of Banking & Finance*, *28*(12), 3037–3054.